Cognitive Computing for Brain–Computer Interface-Based Computational Social Digital Twins Systems

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***Abstract*—To accurately and effectively analyze electroencephalogram (EEG) with high complexity, large amount of data, and strong uncertainty, brain–computer interface (BCI) cognitive computing and its signal analysis algorithms are studied based on the digital twins (DTs) cognitive computing platform. To avoid the influence of noise on EEG analysis results, it is necessary to use filtering and defalsification methods to process EEG. Four methods, including Butterworth filter, finite impulse response (FIR) filter, elliptic filter, and wavelet decomposition, are summarized. Based on the Riemann manifold theory, a feature extraction algorithm under transfer learning based on tangent space selection (TL-TSS) is proposed. In the process of decoding EEG, an EEG decoding method combining entropy measure and singular spectrum analysis (SSA) is proposed. An algorithm performance is tested on the motor imagery dataset of the two International BCI Competitions. It is found that when the training sample size accounts for 5%, the TL-TSS algorithm proposed in this work is superior to other algorithms in classification accuracy. In particular, compared with common spatial pattern (CSP) algorithm, it has great advantages. The classification accuracy of A2, A4, A8, and A9 users is the best, and especially for A8 users, the classification accuracy reaches 97.88%. In summary, in the EEG interface technology of DT cognitive computing platform, the combination of cognitive computing and deep learning can improve the recognition and analysis effect of EEG, which is of great value for further optimization of DT cognitive computing system.**

***Index Terms*—Artificial intelligence, brain–computer interface, cognitive computing, deep learning, digital twins (DTs) system.**

I. Introduction

As SOCIETY'S assets and systems become more complex, the way they are developed, managed, and maintained must evolve. New measures are needed to meet the new challenges posed by digital reinvention. The birth of digital twins (DTs) allows us to recognize what is happening or what is likely to happen, as well as present and future tangible assets [1], [2], [3]. In industrial manufacturing, DT can help engineers and operators understand the performance of the product while obtaining the expected performance of the product. A range of intelligent predictions can be made by analyzing data from connected sensors and combining it with other sources of information. DT computing aims to go beyond the limitations of traditional communication technology to easily create diverse cyberspace by applying high-precision digital information that reflects real-world objects such as things, people, and societies. It not only realizes fast and deep communication but also realizes large-scale, high-precision prediction, and simulation of the future, speeding up the creation of an intelligent society. Large-scale DT bodies are used to reproduce natural conditions or social changes on a global scale to create high-precision prediction models. These simulations and projections are compatible with the principles of sustainable development, and the simulation of energy balance will contribute to the development of ecological space.

Generally, DT is to copy the physical body or system in the real world to the virtual space to generate a clone, and the two eventually form the DTs [4], [5]. Brain is a complex dynamic system composed of different functional subregions. Neural networks distributed in multiple brain regions can realize brain functions. Modeling and simulation reveal the basic working principles of the brain, which can effectively connect neurobiological behavior and cognition, and play an important role in exploring brain functional mechanisms. On this basis, scientists have established a DT cognitive computing platform by fusing multimodal neural imaging data to achieve accurate simulation of large-scale brain dynamics at the level of functional atlas [6], [7], [8]. The mechanism of steady-state visual evoked potential (SSVEP) response was studied based on this model. The establishment of high-precision DT brain model at the level of brain system can not only integrate the results of various biological brain studies but also transform the cross section brain of anatomic biological study into a vivid dynamic brain and working brain. DT cognitive computing platform is expected to play an important role in the research of brain–computer interface (BCI) and attention mechanism.

BCI covers neuroscience, psychology, computer science, biomedical engineering, clinical medicine, and other disciplines. Researchers need to master programming ability and understand biomedical, data processing, and pattern recognition knowledge, which is a comprehensive research field [9], [10]. BCI is a combination of human thinking and machine emerging technology. Through the fusion of human brain and

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machine, the current way of interaction between human and machine, and human and environment can be broken so that humans can break through the limitations of bodies and tools. By establishing a direct communication channel between the brain and the outside world, BCI connects the brain with external devices and establishes a two-way communication interface between the brain and the outside world. In this two-way communication interface as a whole, on the one hand, the brain has to give instructions to control external devices. On the other hand, external devices also need to continuously send various types of feedback information to the brain so that the brain can adjust the control strategy in time and maintain the stability of the whole system [11].

BCI is regarded as an important technical means to combine human intelligence and machine intelligence in the future. In the past two decades, many national laboratories around the world have been conducting research on BCI, realizing a variety of schemes and application modes. A complete set of BCI system mainly includes four links: acquisition and amplification of brainwave, classification, and recognition of brainwave by algorithm, sending recognized instructions to the device and feedback of the device to itself. The brain generates signals to control the user's intentions, and the BCI system can translate these signals into controllable external output commands, which can be used to control external devices.

Cognition has uncertainty, which mainly comes from the uncertainty of the information of objective things, which then directly affects the concept formed by the information mapped to the human brain, so the decision-making of external things also has uncertainty. To effectively process a large number of disordered electroencephalogram (EEG) data, cognitive computing based on BCI is proposed. Based on the preliminary exploration of DT cognitive computing platform, this study focuses on analyzing cognitive computing methods based on BCI and accurate analysis methods of EEG, to improve the efficiency and accuracy of EEG recognition through systematic discussion.

II. Recent Related Works

DT, as a new and rapidly developing technology, is one of the most promising technologies in the field of intelligent design and manufacturing and has attracted extensive attention worldwide. With the application of DT, product performance evaluation has entered the data-driven era. However, the traditional evaluation methods mainly focus on the structural analysis of DT manufacturing and service stage, which fail to integrate multisource information and consider the emotional response of high level in the design stage. To address these problems, Feng *et al.* [12] proposed a DT-based driving method to evaluate product design schemes. First, EEG data, body data, and emotional feedback of human body were obtained. In the evaluation process, human psychological needs, subjective and objective evaluation indexes, and other human factors were systematically considered. A novel EEG analysis based on machine learning was realized. de Kerckhove [13] considered the ubiquitous distribution characteristics of digital information and the advantages of machine learning in developing life logs. In the psychological effect of digital transformation, the externalization of cognitive abilities, such as memory, planning, and judgment, makes human decision-making process transfer to the digital function. By integrating BCI into human body, it can realize the reproduction and modeling of human EEG data and then guide human decision-making.

In terms of DT model recognition, BCI can distinguish, classify, and retrieve information. In terms of decision-making, BCI can evaluate and match numerical objects. On the control side, BCI can accomplish performance generation, design and motion optimization, and operation automation. Cao [14] studied the specific application of BCI based on EEG, including the application of deep learning model to monitor and feedback human cognitive state. Papanastasiou *et al.* [15] explored the paradigm shift of BCI and found that BCI devices had positive effects on people's attention, working memory, and other skills (such as social interaction, imagination, and emotion). Pan *et al.* [16] carried out BCI experiments based on the EEG analysis in people with consciousness disorders and further performed tasks by using EEG to reflect users' intention commands.

As a key bridge between modern neuroscience and brain-like intelligence, computational neuroscience technologies represented by cross-scale brain simulation are helping to uncover the workings of the brain and develop brain-like intelligence and general artificial intelligence algorithms. The level of neurons simulating the brain requires weighing all the nuances of one's own and one's partners. Faced with such a huge scientific problem, the processing power of computers is still limited and cannot be realized in a practical time. Therefore, in-depth research is performed on BCI cognitive computing and related algorithms of signal analysis based on the DT cognitive computing platform so that EEG with high complexity, large amount of data, and strong uncertainty can be analyzed accurately and effectively.

III. DT Cognitive Computing With Brain–Computer Interface

A. DT Cognitive Computing Platform Architecture

DT is summarized as a digital copy of a physical entity in a virtual space in a real physical space, and the two parts communicate with each other. DT simulates, verifies, optimizes, evaluates, and advises, predicts, and controls real entities for people to make decisions, improve performance, and extend the life cycle of physical entities. The ideal DT model involves geometric model, physical model, behavior model, rule model, and other multidimensional, multitemporal, and multiscale models. The DT model is expected to have the characteristics of high fidelity, high reliability, and high precision, so as to truly depict the physical world. From the data dimension, data are the core driving force of DT. Data not only include the whole factor/whole process/whole business-related data throughout the whole product life cycle but also emphasizes the integration of data, such as information physical virtual-real fusion and multisource

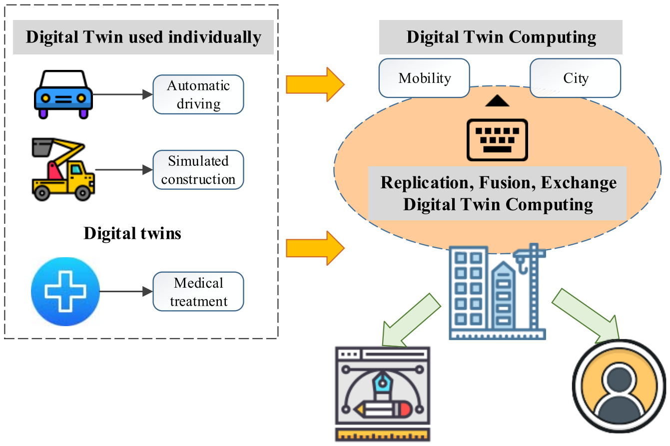

Fig. 1. Overall architecture of DT computing.

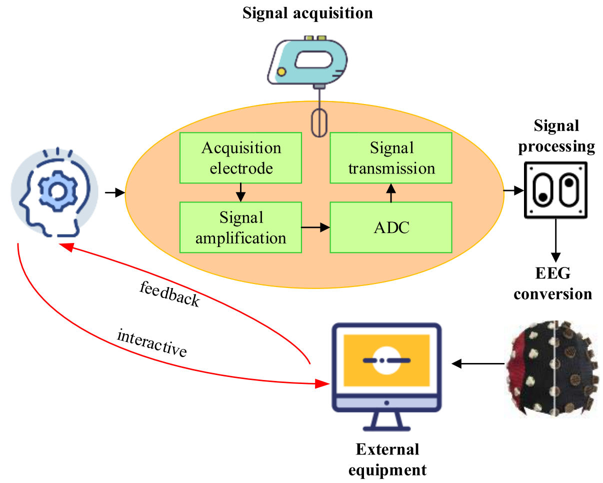

Fig. 2. Schematic of BCI system composition.

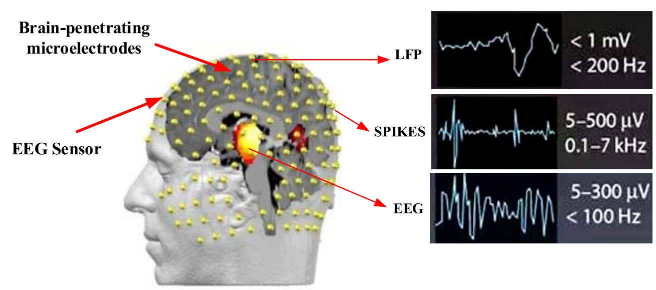

Fig. 3. Signal strength obtained by different interface types.

heterogeneous fusion. The overall architecture of DT computing is shown in Fig. 1.

An enhanced DT is defined as a complex system that interacts not only with its real entities but also with its surroundings and other DTS. The enhanced DT system includes digital counterparts and their surroundings, relationships with other DTS, physical entities and their surroundings, and relationships with other physical entities [17], [18], [19]. Human beings are alive, as opposed to inanimate physical entities. It is intended to apply the enhanced DT system to human beings and then build DT cognitive computing platform. Brain is a complex dynamic system composed of different functional subregions. Neural networks distributed in multiple brain regions can realize brain functions. Modeling and simulation reveal the basic workings of the brain and can effectively link neurobiological behavior and cognition. Periodic visual stimulation can induce SSVEPs distributed in multiple brain regions. Due to its high signal-to-noise ratio, stable spectrum, and high recognition rate, this signal has been widely applied in the fields of BCI, brain cognition, and brain diseases [20], [21]. However, the dynamics of SSVEP at the whole brain level are not fully understood. SSVEP response and spatial distribution characteristics, especially α band stimulation, can stimulate the strongest SSVEP response. This frequency sensitivity is caused by nonlinear entrainment and resonance and is modulated by endogenous factors in the brain.

B. Cognitive Computing Method Based on BCI

BCI is essentially a strategy for control directly via EEG. The control is finally achieved by extracting the characteristics of EEG in different behavior states. The schematic of the BCI system is shown in Fig. 2. As the most important part of the brain, the cortex processes auditory, visual, and sensory information, as well as language, movement, thinking, planning, and personality. Different types of BCI require to pick up signals from different areas of the cerebral cortex. The signal strength obtained by different interface types varies greatly (see Fig. 3). The electrodes for EEG collection are usually placed using the international standard lead 10–20 system.

The use of cognitive computing has improved the ability of DT and the level of science and technology [22], [23]. Natural language processing, machine learning, object/visual recognition, acoustic analysis, and signal processing are just a few of the techniques and techniques that enhance traditional engineering skills. Cognitive DT allows us to design and improve future machines beyond human intuition because it is not only related to what is being built but also for whom [24].

EEG is weak and has strong randomness under the influence of environmental changes and subjects' own factors. The noise frequency of external interference is high and can be separated by a filter. The EEG collected by a noninvasive method contains noise and artifact signals, and the EEG can be processed by filtering and de-artifact methods to avoid affecting subsequent selection and classification. Four methods, including Butterworth filter, finite impulse response (FIR) filter, elliptic filter, and wavelet decomposition, are summarized [25], [26], [27].

1) *Butterworth Filter:* The attenuation rate of the first-order Butterworth filter is 6 dB per octave and 20 dB per ten octaves. The higher the order of the filter is, the faster the amplitude decays in the stopband. The amplitude square function of the Butterworth approximation is expressed as the following

equation:

$$|H\alpha(j\Omega)|^2 = \frac{1}{1+\left(\frac{\Omega}{\Omega_c}\right)^{2N}}. \tag{1}$$

In (1), N is the order of the filter and is a positive integer. Ω_c is the cutoff frequency, and when $\Omega = \Omega_c$, the following equation is obtained:

$$|H\alpha(j\Omega)|^2 = \frac{1}{2} \tag{2}$$

$$H\alpha(j\Omega) = \frac{1}{\sqrt{2}} \tag{3}$$

$$\delta_1 = 20\lg\left|\frac{H\alpha(j\Omega)}{H\alpha(j\Omega_c)}\right| = 3db. \tag{4}$$

When $\Omega = \Omega_c$, all curve characteristics pass 3 db regardless of any value of N.

2) FIR filter is a finite length unit impulse response filter with nonrecursive structure and no output to input feedback. The difference equation of FIR filter is expressed as the following equation:

$$y(n) = \sum_{k=0}^{N-1} b_k x(n-k). \tag{5}$$

The system function is the following equation:

$$H(z) = \sum_{k=0}^{N-1} b_k z^{-k}. \tag{6}$$

In this search, the filtering mode function and spectral function are shown as follows:

$$\omega_{Hm}(n) = \left[0.54 - 0.46\cos\frac{2\pi n}{N-1}\right]R_N(n) \tag{7}$$

$$W_{Hm}\left(e^{j\omega}\right) = 0.54W_R\left(e^{j\omega}\right) - 0.23W_R\left(e^{j\left(\omega-\frac{2\pi n}{N-1}\right)}\right) - 0.23W_R\left(e^{j\left(\omega+\frac{2\pi n}{N-1}\right)}\right). \tag{8}$$

The amplitude function is expressed as the following equation:

$$W_{Hm}(\omega) = 0.54W_R(\omega) - 0.23W_R\left(\omega - \frac{2\pi n}{N-1}\right) + 0.23W_R\left(\omega + \frac{2\pi n}{N-1}\right). \tag{9}$$

3) The elliptic filter is a filter in the passband and stopband ripples, and its advantage lies in the minimum passband and stopband fluctuations. The square amplitude response of the filter is expressed as the following equation:

$$J(\Omega) = |H_\alpha(j\Omega)|^2 = \frac{1}{1+\varepsilon^2 U_N^2\left(\frac{\Omega}{\Omega_c}\right)}. \tag{10}$$

In (10), N is the order and ε is the passband fluctuation. The order of the elliptic filter is the following equation:

$$N = \frac{K(k)K\sqrt{1-k_1^2}}{K(k_1)K\sqrt{1-k^2}}. \tag{11}$$

In (11), $K(x)$ is the complete elliptic integral.

The frequency amplitude characteristics of the elliptical filter are shown in Fig. 4.

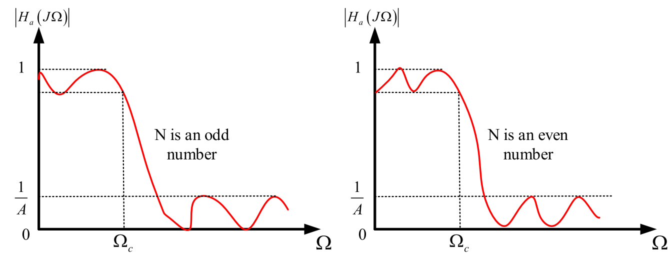

Fig. 4. Frequency amplitude characteristics of elliptic filter.

4) *Wavelet Decomposition:* EEG is processed by wavelet decomposition, and wavelet coefficients and thresholds are used to reconstruct the wavelet and eliminate the corresponding frequency bands of noise signals. First, it is assumed that the function is the following equation:

$$\Phi(t) = L^2(R). \tag{12}$$

This function represents the linear space of square-integrable functions on all real number lines, satisfying the following equation:

$$\int_{-\infty}^{+\infty} \Phi(t)dt = 0 \tag{13}$$

$$\int_R |\psi(\omega)|^2|\omega|^{-1}d\omega < +\infty. \tag{14}$$

In the above equation, $\psi(t)$ is the Fourier transform of $\Phi(t)$ and $\Phi(t)$ is the generating function of the corresponding wavelet. The principle of wavelet decomposition is decomposing the low-frequency part twice, getting the decomposition coefficient sequence, then reconstructing the source signal, and finally getting the signal after filtering the noise. The signal S is reconstructed into the following equation:

$$S = A1 + AAD3 + DAD3 + DD2. \tag{15}$$

For people, the information processing related to cognitive activities includes the information processing process of receiving, detecting, converting, reducing, synthesizing, coding, storing, extracting, rebuilding, concept formation, judgment, and problem-solving of sensory signals. Cognitive computing systems have excellent learning ability. Through evidence-based learning ability, they can quickly extract key information from big data and learn and recognize like human beings. The Bayesian cognitive model effectively combines prior knowledge about the world with sensory evidence to generate a posterior distribution to predict or guide behavior. Assume that data sample C contains n-dimensional features (attribute values), which can be divided into m classes. Given a sample X of an unknown class and the unknown sample is assigned to C_i, the following equation is obtained:

$$P(C_i|X) > P\left(C_j|X\right), 1 \le j \le m. \tag{16}$$

The method is used to classify EEG samples. First, the probability that the samples to be classified belong to each known category is calculated, and then, the maximum probability is selected as the category of the samples according to the probability of the situation.

C. Motion Image BCI Feature Extraction Algorithm

The key to feature extraction of EEG is determining its feature vector. For the feature extraction of multitype motor imagery from EEG, the main method is recognizing the features of the original EEG data and then distinguishing them. In this study, to reduce the calibration time of BCI users, a transfer learning (TL) EEG feature extraction algorithm based on Riemann tangent space (TS) is proposed. A basic rule in Riemannian geometry is that any two lines in the same plane have a common point (intersection), which means that parallel lines are not recognized in the Riemannian geometry. Another stipulation of Riemannian geometry is that a line can be extended indefinitely, but its total length is finite [28], [29]. Multiple BCI devices can connect to the cloud server through wireless connection, and the cloud server is constantly receiving user data sent by the BCI device.

Based on the Riemannian manifold theory, TS is the linear space formed by tangent vectors at a point on a Riemannian manifold. The unsupervised method is adopted to mine sample characteristics, and a single-user EEG feature extraction algorithm can be realized. E_i refers to the EEG after the ith bandpass filtering in the N motion imagination, including the EEG with n leads at t moments. It can get the covariance matrix P_i

$$P_i = \left(E_i E_i^T\right)/(t-1). \tag{17}$$

The n-order covariance matrix $n \times (n+1)/2$ representing N experiments can be represented by N points on a Verriemannian manifold. When the number of leads is large, the dimension of the Riemannian manifold increases sharply. The shortest smooth curve between two points in a Riemannian manifold space is called a geodesic. Logarithmic mapping and exponential mapping are two important reciprocal mapping operators of TSs in Riemannian manifolds. The inner product operation of tangent vectors S_1 and S_2 is defined on the TS $T_p(M)$ to measure the Riemann distance and is expressed as the following equation:

$$\langle S_1, S_2\rangle_p = trace\left(S_1 P^{-1} S_2 P^{-1}\right). \tag{18}$$

In (18), $trace(\cdot)$ represents the trace of the matrix. On a Riemannian manifold M, the logarithmic map of a point P_1 in a TS $T_p(M)$ and the exponential map of a tangent vector S_1 are calculated as shown in the following equations, respectively:

$$\log_p P_1 = S_1 = P^{1/2} \ln\left(P^{-1/2} P_1 P^{-1/2}\right) P^{-1/2} \tag{19}$$

$$\exp_p S_1 = P_1 = P^{1/2} \exp\left(P^{-1/2} S_1 P^{-1/2}\right) P^{-1/2}. \tag{20}$$

Furthermore, the geodesic distance between points P and P_i on the Riemannian manifold M is obtained as follows:

$$\delta(P, P_i) = \sqrt{\left\langle \log_p P_i, \log_p P_i\right\rangle_p} = \left\|\ln\left(P^{-1} P_i\right)\right\|_F. \tag{21}$$

The geodesic distance has the important property as follows:

$$\delta\left(W^T P_1 W, W^T P_2 W\right) = \delta(P_1, P_2). \tag{22}$$

It is revealed that after affine transformation of points P_1 and P_2 and their corresponding covariance matrix, the distance of Riemannian geodesic between two points remains unchanged.

Similar to the TS algorithm, the feature extraction algorithm TL based on TS calculates the Riemann mean of the covariance matrix corresponding to the EEG in each session and uses it as a reference point [30], [31]. The TL-TS algorithm inputs the set of feature tangent vectors and their categories into the classifier for training. The Riemannian mean of the test session is used as the reference point of the test stage, and the generated feature tangent vector of the test sample is input into the trained classifier for recognition. Although the TL-TS algorithm realizes signal transfer across users, its flexibility still needs to be improved due to the large difference in EEG between BCI subjects. Therefore, a feature extraction algorithm based on TL based on TS selection (TSS) is proposed. The core is a floating search strategy, repeatedly moving forward and backward to avoid falling into local optima.

The TL-TSS algorithm trains the sample feature tangent vector set for a specific target user in each round of comparison. First, a best source user feature tangent vector set is selected forward from the current source user feature tangent vector set, and then, a worst source user feature tangent vector set is deleted backward. When the proportion of training samples of target users is less than 50%, the TL-TSS algorithm adopts the forward user selection strategy. As long as the classification accuracy of the training sample of the target user is higher than the previous optimal value, it can be added into the feature tangent vector set of the optimal source user.

D. EEG Decoding Method Based on Deep Learning

EEG decoding is mainly used to separate task-related components from EEG and screen out task-related brain activity information to identify different cognitive tasks in the brain. Signal decoding refers to completing the signal acquisition and data preprocessing. Second, it performs signal analysis and feature extraction. Finally, there is the learning algorithm for EEG pattern classification [32], [33], [34]. In this section, the EEG decoding method based on deep learning is studied.

At present, EEG decoding based on entropy measure mainly reconstructs EEG through various signal processing and then uses an entropy measure method to further optimize feature extraction [35]. On this basis, an EEG decoding method combining entropy measure and singular spectrum analysis (SSA) is proposed. The overall system framework is shown in Fig. 5. SSA mainly includes four steps: embedding, decomposition, grouping, and reconstruction. In the embedding step, the phase space reconstruction method is used to map one-dimensional time series signals to obtain the trajectory matrix. In the decomposition step, singular value decomposition is performed on the trajectory matrix X, which is decomposed into w elementary matrices in the form of the following equation:

$$X_j = \sqrt{\lambda_j} U_j V_j^T \tag{23}$$

$$X = X_1 + X_2 + \cdots + X_w = \sum_{j=1}^{w} \sqrt{\lambda_j} U_j V_j^T. \tag{24}$$

In the above equation, λ_j and U_j represent the eigenvalues and eigenvectors of the covariance matrix XX^T, respectively,

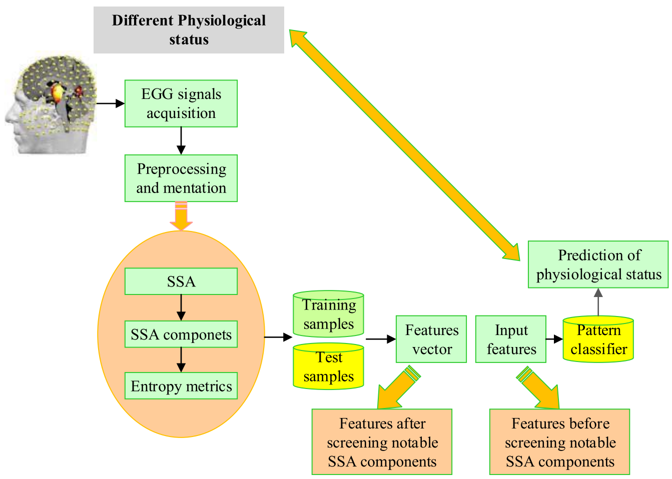

Fig. 5. EEG decoding system for entropy measure and SSA.

and U_j and V_j represent the left and right singular vectors, respectively.

In the reconstruction steps, grouping and diagonal averaging are included. The grouping process needs to form the subscript set based on the result of singular value decomposition and divide its subset to obtain v subsets. The grouping matrix X_{c_k} is calculated and combined into a new result matrix $\widetilde{X}$ as shown in the following equation:

$$\widetilde{X} = \sum_{k=1}^{v} X_{c_k}. \tag{25}$$

Diagonal averaging is performed on the diagonal $i + j = n + 1$ of $\widetilde{X}$, and all diagonal elements are averaged to obtain one-dimensional time series. The time series signal $t(n)$ is expressed as the sum of the one-dimensional time series $\underset{c_k}{\widetilde{y}}(n)$ reconstructed by each order, as shown in the following equation:

$$t(n) = \sum_{c_k=1}^{w} \sum_{n=1}^{N} \underset{c_k}{\widetilde{y}}(n). \tag{26}$$

By decomposition and reconstruction, EEG is expressed as the sum of the singular spectral analysis components as shown in the following equation:

$$s(n) = \sum_{p=1}^{w} \sum_{n=1}^{N} \underset{p}{\widetilde{s}}(n). \tag{27}$$

In this study, the support vector machine (SVM) of machine learning method is used to establish a pattern classifier in the EEG decoding algorithm. The sample dataset composed of the eigenvector x_i and label y_i of the sample is the following equation:

$$\left\{(x_i, y_i) | x_i \in R^n;\ y_i \in \{-1, +1\}\right\}. \tag{28}$$

To maximize the spacing between data samples containing different label classes, a classification hyperplane should be constructed. The distance from the sample point to the classified hyperplane is expressed as the following equation:

$$d(x_i, H) = \frac{|\langle w, x_i \rangle + b|}{\|w\|}. \tag{29}$$

Interval distance maximization is converted to a minimum problem, as in the following equation:

$$\min_w \frac{\|w\|}{2}, \text{ subject to : } y\left(\left\langle w^T x_i \right\rangle + b\right) \geq 1. \tag{30}$$

In the process of modeling, points far from the classification hyperplane do not have an obvious influence on the position of the classification hyperplane. SVM seeks to minimize the empirical risk and confidence range to optimize the generalization performance of the model with limited sample information.

E. Case Verification

The motor imagination datasets of two International BCI Competitions (third and fourth) are selected as the experimental data. Common spatial pattern (CSP) algorithm and the TS algorithm are selected for single-user feature extraction. General learning regularized CSP (GLRCSP) algorithm and the TL-TS algorithm are selected for the general-user feature extraction algorithm. Weighted Tikhonov regularized CSP (WTRCSP) and the TL-TSS algorithm proposed in this research are selected as the specific-user feature extraction algorithm. The effectiveness of TL is verified by an example, and the performance difference between TS and CSP-based methods in feature extraction is compared. The hardware and software platform of the experiment are given as follows. For hardware, the processor is Intel Core I7 8700, CPU3.2 GHz, 64 GB of memory, and Software is Win 10, MATLAB 2016a.

The number of leads in dataset IVa of the third BCI competition is 118. To avoid dimensional explosion, only the 42nd to 66th leads related to motion imagination are extracted. The datasets of the two BCI competitions are collected from 0.5 to 2.0 s after visual cues. All algorithms use linear discriminant analysis (LDA) to better compare the performance of feature extraction algorithms. Five users in the third competition's dataset include al, aw, and ay with good performance, while aa and av are with poor performance. The nine users in the third competition's dataset include the high performers A1, A3, A7, A8, and A9, while A2, A4, A5, and A6 are with poor performance.

An EEG recognition task of eye state is designed to compare the performance of EEG decoding methods. The EEG data samples selected are public EEG datasets provided by the Epilepsy Research Center, University of Bonn, Germany. This experimental study uses EEG data of two subsets (A and B), including EEG data of five healthy subjects under different states (open and closed eyes), with a sampling frequency of 173.61 Hz. As for the design of contrast experiment, EEG entropy measure features are extracted directly from EEG, and there is no signal processing method integrating SSA decomposition and reconstruction in the process of EEG entropy feature extraction.

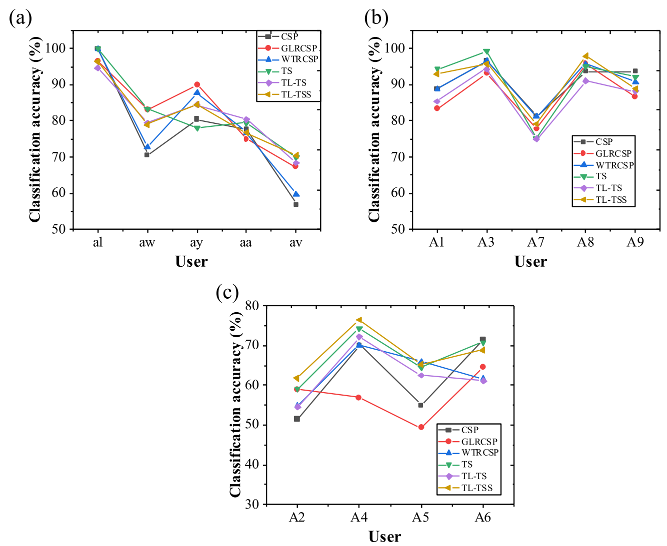

Fig. 6. Classification accuracy of different users in datasets of two competitions. (A) Users in the third session, (B) users with good performance in the fourth session, and (C) users with poor performance in the fourth session.

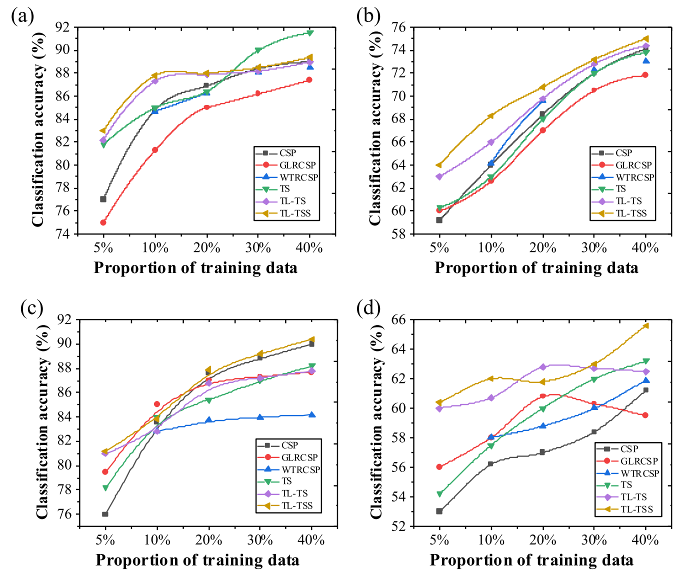

Fig. 7. Influence of different proportions of training data on classification accuracy. (A) Users with good performance in the third session, (B) users with poor performance in the third session, (C) users with good performance in the fourth session, and (D) users with poor performance in the fourth session.

IV. Results and Discussion

A. BCI Data Feature Extraction Results Based on Riemann TS

Based on EEG training datasets of two BCI competitions, the classification accuracy of different user test samples obtained is shown in Fig. 6. In the dataset of the third competition, the classification accuracy of the TS algorithm is generally good, especially for al users (including 224 samples), the classification accuracy reaches 100%. Also, the classification accuracy of aw users (including 56 samples) is also the highest among several comparison algorithms. For ay users with only a small number of training samples, the GLRCSP algorithm shows the optimal classification effect, and the classification accuracy reaches 89.92%. For users with poor performance, users aa and av have the best classification performance in the TL-TS and TL-TSS algorithms, respectively. In the fourth contest dataset, the number of users increases and individual differences are more obvious. As the TL-TS and GLRCSP algorithms directly integrate all user feature tangent vectors, the classification accuracy is the lowest. The proposed TL-TSS algorithm achieves the best classification accuracy for A2, A4, A8, and A9 users, especially for A8 users, which reaches 97.88%.

The classification accuracy of different algorithms for user EEG in the case of different proportions of training samples is analyzed, as shown in Fig. 7. The EEG classification accuracy of different algorithms is compared when the proportion of training data is 5%, 10%, 20%, 30%, and 40%. It should be noted that when the proportion of training samples is 5%, the WTRCSP algorithm cannot be cross-verified, so it cannot run normally. As shown from the results, for users with good performance, when the training sample size accounts for 5%, the proposed TL-TSS algorithm is superior to other algorithms in classification accuracy. In particular, compared with the CSP algorithm, it has great advantages. With the increasing proportion of training sample size, the gap between the CSP algorithm and other algorithms is gradually narrowed and gradually shows its superiority. Therefore, for users with good performance, only when the training sample size is small, can the TL algorithm give full play to its classification advantages. For users with poor performance, the corresponding algorithm based on TS is superior to the corresponding algorithm based on CSP in the overall classification accuracy.

The above analysis shows that the proposed TL-TSS algorithm has a higher accuracy of EEG classification for users with poor performance. Therefore, the classification effects of different algorithms in the six users with poor performance in the datasets of the two competitions are analyzed, and the specific results are shown in Fig. 8. For users with poor performance, the overall performance of feature extraction algorithm based on TL is better than that of single-user feature extraction algorithm. The feature extraction accuracy of general user and specific user based on the TS algorithm is higher than that based on CSP. As the number of training samples increases, the differences among users gradually appear, and the TL-TS algorithm cannot adapt to the bande training samples. Therefore, compared with the proposed TL-TSS algorithm, the gap in classification performance increases with the increase of training sample size.

B. Comparison of Performance of EEG Decoding Methods

For the EEG data samples used in the experimental evaluation, the SSA analysis method is first adopted to decompose each single-channel time series sample, so as to obtain the ten-order SSA component. Next, four kinds of entropy features (approximate entropy, sample entropy, fuzzy entropy, and multiscale entropy) are extracted from the SSA components of order 1–10, and the corresponding statistical results of entropy

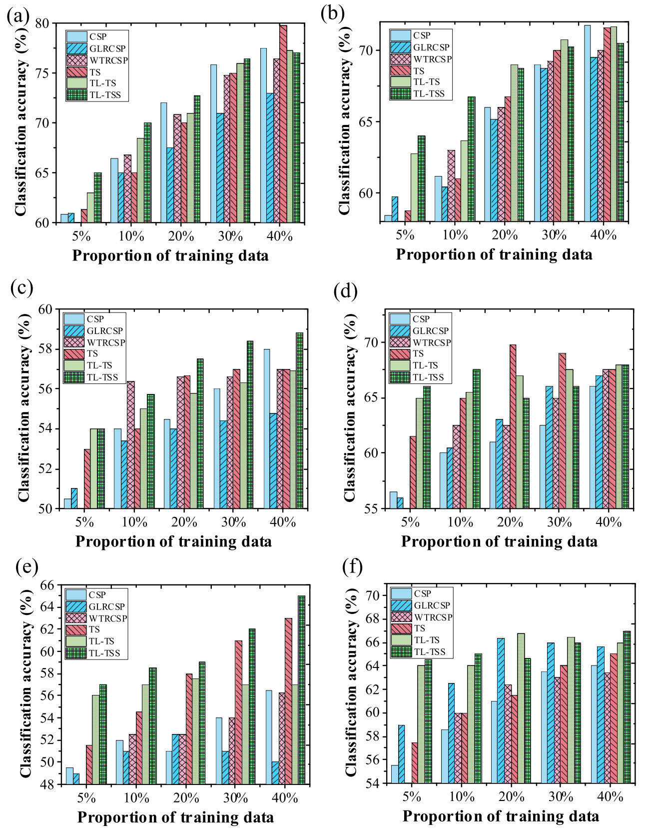

Fig. 8. Classification accuracy of different algorithms in users with poor performance (A) aa user, (B) av users, (C) A2 user, (D) A4 user, (E) A5 user, and (F) A6 user.

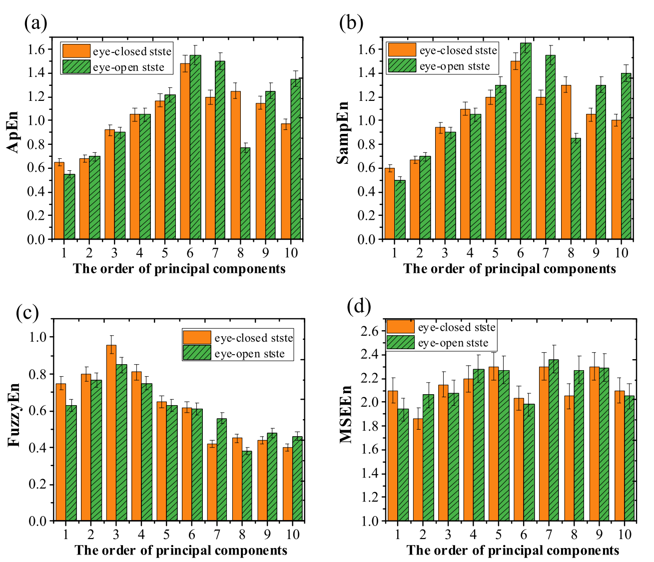

Fig. 9. Characteristics of EEG SSA component entropy measure. (A) Approximate entropy, (B) sample entropy, (C) fuzzy entropy, and (D) multiscale entropy.

are shown in Fig. 9. It is found that there are significant differences in approximate entropy and sample entropy between closed and open eyes. For approximate entropy and sample entropy, the first five points of entropy of SSA component of order EEG1–10 are 1, 5, 10, 9, and 8. For fuzzy entropy, the first five entropy characteristics of SSA components of order EEG1–10 are 1, 8, 3, 5, and 9. For multiscale entropy, the first five are 3, 2, 6, 4, and 9.

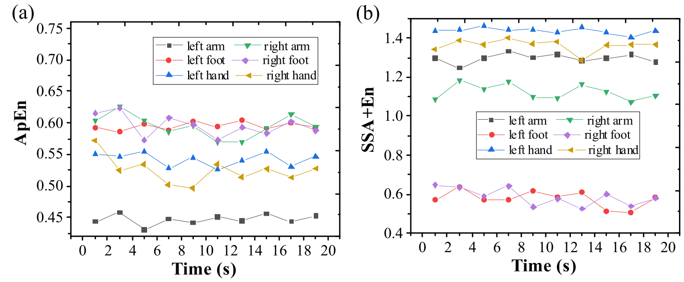

Fig. 10. EEG feature extraction results. (A) Approximate entropy and (B) SSA binding entropy measure.

The purpose of approximate entropy is calculating the relationship between the model and the new model when the dimension of the model changes. An approximate entropy algorithm with a window size of 1500 is employed, and six types of imagined motion EEG are calculated [see Fig. 10(A)]. The EEG is decomposed and reconstructed by SSA combined with entropy measure, and the results such as EEG feature extraction are obtained [see Fig. 10(B)]. There are few intersections between single curves, so we can distinguish different kinds of imagination movement clearly. Also, a certain kind of curve itself has little fluctuation and high stability. Compared with the approximate entropy method, SSA combined with entropy measure has more advantages in EEG feature extraction.

V. Conclusion

DT has been widely studied and applied by experts, scholars, and enterprises in various industries, backgrounds, and levels around the world. At present, DT has carried out more application exploration and landing practice in the field of manufacturing and has broad application prospects in aerospace, electric power, automobile, smart city, health care, and other fields. Based on the DT cognitive computing platform and BCI, this research explores the application of cognitive computing methods in EEG recognition and analysis. A feature extraction algorithm based on TL-TSS is proposed. The core of the algorithm is floating search strategy, which repeatedly moves forward and backward to avoid falling into local optimum. The proposed algorithm shows high classification accuracy on the motor imagination dataset of the International BCI Competition.

In addition, we innovatively propose an EEG decoding method combining entropy measure with SSA. The results show that this method can optimize the decoding performance of EEG and has a broad application prospect in the fields of physiological signal analysis and prediction and cognitive computing. Because the semisupervised learning algorithm is used in this study to analyze EEG offline, it fails to meet the requirements of real-time signal analysis in practical applications. In the following research, we will try to use a semisupervised learning algorithm to perform the online analysis on motor imagery EEG and continuously update the classifier to achieve higher classification accuracy.

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